

Manuel Capiendo

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EXPERIENCE

Universal Destinations & Experiences | Technical Solutions Architect Intern Sep. 2026 – Present

- Building an internal SWE-bench-style benchmark to evaluate AI coding agents on company codebases, informing adoption decisions for AI-assisted development across engineering teams.
- Supporting solutions architects in evaluating emerging technology, including computer vision, AI, and robotics, for use across Universal's parks and business units.

NextEra Energy | Software Development Engineer Intern May 2026 – Aug. 2026

- Built an ETL pipeline that extracts asset and condition detections from multiple AI models, normalizes their labels into a unified schema, and loads them into PostgreSQL and OpenSearch with update and delete handling.
- Built the corresponding filtering and search UI in React and TypeScript, including advanced filter menus and shareable deep links, with unit and end-to-end tests to validate functionality.

Universal Destinations & Experiences | Software Quality Engineering Intern Jan. 2026 – Apr. 2026

- Extended mobile automation frameworks in Java and Python (Selenium, Appium) to validate app releases for Orlando, California, and Japan properties and catch regressions before production.
- Developed and executed test plans for web and eCommerce platforms, including cross-browser and cross-device validation with LambdaTest across multiple release cycles.

NextEra Energy | Programmer Analyst Intern Sep. 2025 – Dec. 2025

- Built an AI-powered API discovery tool (Angular, Flask, OpenAI) that matches natural language requests to relevant APIs across a catalog of 18,000+ internal nuclear APIs.
- Led a front-end architecture refactor and presented the API discovery tool to the director and nuclear development team, who adopted it as the foundation for nuclear IT development standards.

NextEra Energy | Nuclear Solutions Analyst Intern May 2025 – Aug. 2025

- Built autonomous inspection routes for robotic quadrupeds at the St. Lucie Nuclear Power Plant to automate facility monitoring, and presented the project to regional Nuclear Regulatory Commission directors.
- Built a VR control room prototype in Unity for Meta Quest using LiDAR-scanned 3D environments to evaluate remote operator training as an alternative to on-site visits.

Sparton Corporation | Acoustic Engineering Intern May 2024 – Aug. 2024

- Identified a sonobuoy production testing bottleneck and coordinated with multiple departments to customize existing impedance analyzer driver code in C, removing the need for an advanced operator.
- Integrated an automated test station with a SQL Server database, reducing test cycle time for piezoelectric ceramic components from 160 to 40 seconds (75%) and restoring full production capacity.

PROJECTS

Grant Compliance Assistant | Senior Design, FAU Aug. 2026 – Dec. 2026

- Extending GOST, an open-source federal grant management platform (Vue, Node/Express, PostgreSQL, Docker), with AI-assisted grant matching, source-linked compliance checklists, and deadline tracking for small local governments.
- Building the deadline tracking subsystem, including Express API endpoints and Vue dashboard and calendar views that compute deadline status at request time from due date, completion, and verification rules.

EDUCATION

Florida Atlantic University Expected Dec. 2026

Bachelor of Science in Computer Science, Minor in AI | GPA: 3.32

Certifications: AWS Certified Cloud Practitioner, Lean Six Sigma Yellow Belt

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL

Frameworks: React, Vue, Angular, Node.js/Express, Flask

AI/ML: PyTorch, scikit-learn, pandas, NumPy, OpenAI API

Data & Search: PostgreSQL, OpenSearch, SQL Server, MongoDB

Cloud & Tools: S3, Lambda, Docker, Linux, Git, GitHub, Jira

Testing: Vitest, Playwright, Selenium, Appium, LambdaTest